

FL-01 Prompt Engineering Assignment

Student: Vikas Kumar Prajapati
Track: Machine Learning Engineering
Project: IntelliPlant

Objective: Improve a weak prompt through five prompt-engineering techniques, compare outputs, and create a reusable prompt template.

Baseline

Field	Details
Technique	Naive Prompt
Prompt Added	Build a machine learning pipeline.
Observed Improvement	Claude asked clarifying questions because the request lacked context.
Remaining Limitation	Prompt too vague.
Next Step	Add Role Assignment.

V1

Field	Details
Technique	Role Assignment
Prompt Added	You are a Senior Machine Learning Engineer.
Observed Improvement	More technical and solution-oriented.
Remaining Limitation	Still generic.
Next Step	Add Context.

V2

Field	Details
Technique	Context & Motivation
Prompt Added	I'm building IntelliPlant for plant health prediction.
Observed Improvement	Recommendations became project-specific.
Remaining Limitation	Still assumed implementation details.
Next Step	Add Few-shot Example.

V3

Field	Details
Technique	Few-shot Example

Prompt Added	Added a small example.
Observed Improvement	Output matched the desired style.
Remaining Limitation	Needed better organization.
Next Step	Add Output Structure.

V4

Field	Details
Technique	Output Structure
Prompt Added	Specified headings.
Observed Improvement	Easy to scan and document.
Remaining Limitation	Reasoning still limited.
Next Step	Add Step Decomposition.

V5

Field	Details
Technique	Step Decomposition
Prompt Added	Asked it to solve the task step by step.
Observed Improvement	Produced a complete engineering plan with reasoning and trade-offs.
Remaining Limitation	Longer response but much more useful.
Next Step	Final template.

Claude vs ChatGPT

Aspect	Claude	ChatGPT
Reasoning	Clarified assumptions before answering.	Answered directly using reasonable assumptions.
Tone	Engineering-focused.	Beginner-friendly.
Strength	Trade-offs and deployment advice.	Clear formatting and concise explanations.
Weakness	More verbose.	May assume missing context.

Reusable Prompt Template

You are a Senior Machine Learning Engineer. I'm building **[PROJECT]** with the goal of **[GOAL]**. Design an end-to-end ML solution using sections for Problem, Dataset, Feature Engineering, Model Selection, Training, Evaluation, Deployment, and Future Improvements. Explain assumptions, reasoning, and trade-offs using Markdown.

Reflection

Each prompt-engineering technique improved the response differently. Context made the biggest impact, while output structure and step decomposition made the final response significantly more useful and reusable.